Detailed Summary of the v6 ETH Report

Multi-Edge Packets, Nonzero Higher Modes, Deeper Recursion, Measured Speedups, Precision Savings, and Full-Algorithm Limits

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Abstract

This report explains the results and limitations of revision 6 of the Ram Calculus investigation of OpenAI's manuscript *The Exponential Time Hypothesis is False*. V6 implements two-real-edge packets with nonzero higher geometric modes and, in a separate test family, three active singleton-packet recursion levels. The final evidence consists of 100 validated worker outputs on five small fixtures: 75 timing runs and 25 diagnostic runs. All final counts agree with independent exact oracles; non-direct workers also reproduce every root local polynomial coefficient, and diagnostic runs check every executed active child target. The combined method has packet source/variant median elapsed-time ratios between 4.72× and 7.58×. On the deeper fixture its median is 22.094 seconds versus 858.467 seconds for the finite source-form comparator, a host-specific ratio of 38.85×; the median CPU-time ratio is 38.06×. Substantial timing variation limits extrapolation. The Boolean packet coefficient precision falls from 485 to 54 bits, with four additional numerator guard bits; the deeper path falls from $5 \to 261 \to 453 \to 581$ to $5 \to 23 \to 36 \to 45$ bits. Process-memory peaks are dominated entirely by pre-call high-water marks, so no algorithm-specific memory saving is established. We explain the exact quotient and recovery identities, failure controls, a further proved but unbenchmarked Boolean reduction, and the work still required for a full-algorithm improvement. This is a detailed summary of existing v6 evidence, not a new benchmark, literature-status audit, or independent verification of the source's full ETH claim.

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1 Principal conclusions and scope

Revision 6 [1] advances the executable investigation beyond the singleton packets and two active levels used previously. Its main contribution is to connect explicit finite-algebra identities to tested multi-port child tensors, nonzero higher modes, and an additional recursive level. The strength of this result depends on stating its scope precisely.

1. **Two-edge packets are genuinely exercised.** Each packet in the new packet family has two real positions, and the generated packet tensors have two ports. Their coefficients can couple both edges.
2. **Higher geometric orders are nonzero and necessary.** They are explicitly summed, not set to zero. Equal moment functions are combined by an exact quotient; this is different from discarding orders.
3. **Three active levels are genuinely exercised.** A separate singleton family follows $4 \to 3 \to 2 \to 1$ edges, with three active levels followed by a leaf. Intermediate tensors and return values need not lie in the scalar subring.
4. **Exact agreement and local speedups are demonstrated.** All completed final-suite outputs pass their exact checks. Timing benefits are observed against the specified finite comparator, with important variability and non-beneficial symmetry cases.
5. **The precision savings are mathematically certified.** They use degree-aware target reconstruction and two-adic denominator valuations, rather than empirical reduction of arithmetic precision.
6. **Memory savings and a full-algorithm improvement are not demonstrated.** The selected memory metric does not resolve algorithm-specific peaks. No improved general SAT exponent follows.

The most important qualification is that the packet and deep-recursion families are *separate*. There is no measured instance combining general multi-edge packets, independently varying higher moments, and three active levels. The packet roots are disconnected across their two packets, although each packet has internal two-edge coupling. The deep root is a factorized, deliberately easy input. Every measured root has only four Boolean edge variables.

Throughout this report, *source comparator* means the implemented **finite, rescaled source-form contraction and Hermite routine**, with its conservative precision schedule. It is not an author-supplied full SAT executable. Furthermore, all packet comparators already share the exact higher-mode coalescence. Their timings therefore do not measure the cost of literal enumeration of every mode tuple in the source paper.

The new v6 section occupies physical PDF pages 6–19 of the 80-page report. V5 and earlier scientific material follows from page 20 onward. Historical tests retained there are not part of the new 100-output suite, and speed ratios from different revisions must not be multiplied into a claimed cumulative full-algorithm speedup.

2 Test families and comparison methods

2.1 Five methods on matched inputs

The five variants have the following roles. All applicable variants use the same input tensors, requested target precision, local polynomial construction, and within-family interpolation schedule.

Method	Meaning
Direct	Exhaustive contraction of the small root, with no active recursion.
S	Source-form Hermite target recovery and conservative working precision.
D	Degree-certified target map and reduced working precision, without symmetry reduction.
G	Certified symmetry reduction, retaining S's precision schedule.
DG	Both degree-aware precision and certified symmetry reduction.

All recursive variants share a bounded exact-key child cache, with capacity 16 per active call. In the packet family the interpolation basis construction is common. In the deeper family the v5 degree/precision basis cache is retained. These choices are part of the measured implementation, not evidence of a general space bound.

2.2 The four packet fixtures

There are two packets, with edges $\{0, 1\}$ and $\{2, 3\}$. Each has a row and a column tensor; their four entries are indexed by two Boolean inputs. The default table $(0, 1, 1, 1)$ is OR. At the root each old vertex forms its own side-pure group. Child networks have two edges and are solved directly at leaf threshold two.

Case	Definition	Exact answer
P01	Identical OR/OR tensors in both packets	9
P02	First row tensor changed to XOR, $(0, 1, 1, 0)$	6
P03	First row tensor identically zero	0
P04	Entry j of vertex i equals $2^{15} + 7i + 3j + 1$	Below
D01	Separate deeper singleton fixture described below	1

P01 has three satisfying assignments in each packet, hence $3^2 = 9$. P02 has two in its modified packet and three in the other, hence six. P03 checks exact propagation of a zero contraction. P04 is a *weighted contraction*, not a Boolean SAT count, with answer

$$18\,482\,797\,547\,933\,526\,180.$$

The unspaced integer stored in the evidence is `18482797547933526180`. It exceeds 2^{64}, but is recovered exactly by arbitrary-precision arithmetic. The Boolean packet cases request $p = 5$ output bits; the weighted case requests $p = 65$. Elementary nonnegative upper bounds are smaller than the output moduli, so agreement identifies the integer answer, not just an ambiguous residue.

2.3 The separate deeper fixture

D01 has four row unaries $(1, 0)$ and four column unaries $(1, 1)$. Only the all-zero edge assignment survives, so the exact answer is one. The groups have sizes $3 + 1$ on each side at the four-edge root,

then $2+1$ for active three-edge children, and singleton groups for active two-edge children. The path is

$$4 \longrightarrow 3 \longrightarrow 2 \longrightarrow 1.$$

The four-, three-, and two-edge calls are active; one-edge calls are leaves. This tests recursive algebra, not difficulty of the root count. A sixth coded candidate, D02, was not part of the measured suite and must not be counted as a validated benchmark.

3 What the multi-edge formulation does

3.1 An exact deformation of the count

The packet family uses $L = 4$, $a = 2$, $u = 1$, local scale $\alpha = 8$, and centers $t \in \{-3, -1, 1, 3\}$. Real packet positions are $x, y \in \{0, 2\}$. Define

$$\mathcal{C}(T) = \prod_{s=-4}^{3} (T - s), \qquad \mathcal{C}'(0) = -144, \qquad K_{xy}(T) = \frac{\mathcal{C}(T)}{-144(T - y + x)}.$$

The apparent division by $T - y + x$ cancels a polynomial factor, since $y - x$ is one of the roots. Each entry therefore has degree seven. At the target, the diagonal entries equal one and the off-diagonal entries equal zero: $K(0) = I$ on the two real positions.

Let $A(Y)$ and $B(X)$ be the products of old row and column tensor entries, and put $\ell_0(z) = 1 - z$, $\ell_1(z) = z$. The deformed contraction is

$$F(T) = \sum_{X,Y} A(Y)B(X) \prod_{P} \prod_{x \in \{0,2\}} \ell_{Y_{P,x}} \left(\sum_{y \in \{0,2\}} K_{xy}(T) X_{P,y} \right). \tag{1}$$

At $T = 0$, the row factors select $Y = X$. Consequently

$$F(0) = \sum_{X} A(X)B(X).$$

For M edges, $\deg F \le 7M$; at the four-edge root the degree bound is 28. This exact target preservation is the reason that computing a deformation and reconstructing its value can recover the original count.

At an odd center, all real-position differences are even and $K_{xy}(t) = 0$. The finite setup places the row and column endpoints in different groups, so the tested real interactions are crossing interactions. Shifting to $T = t + 8\zeta$ activates them. Arbitrary mixtures of local and crossing interactions were not tested.

3.2 Extension arithmetic and full local polynomials

The packet coefficient ring is

$$\mathcal{R}_B = (\mathbb{Z}/2^B\mathbb{Z})[U]/(U^5 + U^2 + 1).$$

Its residue field is $\mathbb{F}_{32}$. There are enough distinct binary residue representatives for 29 interpolation points, and differences of distinct representatives are units. Every executed packet root uses 29

evaluations at each of four centers, hence 116 local evaluations. The complete local polynomial is reconstructed, not just one value.

The use of a fixed five-coordinate extension is sufficient for this degree bound. It is not a proof that one fixed extension suffices for arbitrarily large recursive instances. In the separate deep family, the frozen v5 four-coordinate extension and singleton kernel are used; the two families should not be identified as one ring-size experiment.

4 Higher modes: aggregation is not omission

4.1 Why many nonzero orders share one observation

At $T = t + 8\zeta$, define

$$N = \frac{\mathcal{C}(T)}{\mathcal{C}'(0)}, \qquad A_x = \frac{N}{T+x}, \qquad r_x = \frac{2}{T+x}.$$

The denominators are units. Writing the two column positions as $y = 2j$ with $j \in \{0, 1\}$ gives the geometric expansion

$$K_{x,2j}(T) = A_x \sum_{b \geq 0} r_x^b j^b \pmod{2^B}.$$

For the column-bit observations, with $j^0 = 1$ also at $j = 0$,

$$L_0(X) = X_0 + X_2, \qquad L_b(X) = X_2 \quad (b \geq 1).$$

Thus every positive order has the same *observation function*, although its coefficient need not vanish. Put $L_+(X) = X_2$. Then

$$\sum_y K_{xy}(T) X_y = A_x L_0(X) + B_x L_+(X) \pmod{2^B}, \qquad B_x = \sum_{b=1}^{B-1} A_x r_x^b. \tag{2}$$

Since r_x is divisible by two, sufficiently high terms vanish modulo 2^B. The implementation multiplies and adds every nonvanishing geometric order until the tail is zero. It does not keep only order zero or only order one. This is an exact reduction of redundant observations on the specified finite support.

The independent closed expression

$$B_x = N\big((T+x-2)^{-1} - (T+x)^{-1}\big) = K_{x,2}(T) - K_{x,0}(T) \pmod{2^B}$$

follows by multiplying the geometric sum by $1 - r_x$. Diagnostic workers assert agreement at every interpolation point. The closed value is also computed in timed workers as common overhead; only the equality assertion is diagnostic-only. It is *not* used to replace the geometric loop in the timed comparison.

The original valuation truncation is respected: terms of total source weight at least B vanish modulo 2^B. Identifying equal observation functions commutes with substitution and summation. It does not mean that the original, much larger formal-mode enumeration was executed. Nor does $j^b = j$ hold for arbitrary tile coordinates. A larger tile can have genuinely independent higher moments.

4.2 A verified failure of dropping the higher modes

At $t = -3$, $\zeta = 1$, hence $T = 5$, take $(X_0, X_2) = (0, 1)$ and $(Y_0, Y_2) = (0, 0)$. The full row-factor product is

$$(1 - K_{0,2}(5))(1 - K_{2,2}(5)) = 841 \cdot 505 = 424705.$$

Keeping only the zeroth mode instead gives

$$(1 - K_{0,0}(5))(1 - K_{2,0}(5)) = 505 \cdot 361 = 182305.$$

The control records the difference modulo 2^{40}. It proves that omission changes a local calculation. It is not a test claiming that an invalid algorithm returned a meaningful final SAT count.

4.3 What the counters establish

Every non-direct packet diagnostic has 116 local evaluations, 232 closed-kernel comparisons, and 224 nonzero coefficients of each of orders one and two. Representative higher-order records are:

Family/method	Coefficient bits	Largest order	Orders evaluated	Higher products
P01/S	485	481	107976	2352
P01/D	54	50	11432	2352
P01/G	485	481	107976	1344
P01/DG	54	50	11432	1344
P04/S	545	541	121416	560
P04/DG	114	110	24872	560

Here "higher products" counts nonzero contracted products with a positive higher-mode exponent; it is not a count of independent moment functions. P03 has zero such contracted products because an old tensor vanishes, but it still computes nonzero kernel coefficients of orders one and two. These distinctions prevent mistaking a zero output fixture for a test with inactive higher modes.

5 Two-port children and certified symmetry

5.1 Coefficient extraction produces genuine packet tensors

Replace L_0, L_+ by formal variables Z_0, Z_+. The row polynomial for a packet is

$$\prod_{x \in \{0,2\}} \ell_{Y_x}(A_x Z_0 + B_x Z_+).$$

Its monomial indices satisfy

$$(\beta_0, \beta_+) \in \mathbb{N}^2, \qquad \beta_0 + \beta_+ \le 2,$$

giving six states. The row tensor is the corresponding coefficient, indexed by the two row bits. The column tensor is

$$(X_0 + X_2)^{\beta_0} X_2^{\beta_+},$$

indexed by the two column bits. These tensors are not presumed to factor into unary functions. Group factorization then yields

$$F(t + 8\zeta) = \sum_{\boldsymbol{\beta}} \prod_{g} Z_{g,\boldsymbol{\beta}} \pmod{2^B}.$$

The packet implementation creates four two-edge child networks and solves them at leaf threshold two. This is a real child-construction test, but only one active level is present in this family.

5.2 When 36 assignments can become 21 representatives

Two packets give $6^2 = 36$ coalesced assignments. If the packets are exactly interchangeable, the unordered pairs of six states number $6 \cdot 7/2 = 21$. The code certifies a whole-packet interchange that preserves ordered real positions, tensor entries and induced axes, bipartite sides, creation precisions, and grouping up to a group bijection. A symmetry of the final scalar count alone is insufficient: the entire deformed computation must be invariant.

Orbit aggregation uses

$$\sum_{\beta} Q(\beta) = \sum_{O} |O| Q(\beta_O).$$

There is no division by group order, which is important in a ring where even integers are not units. Arbitrary swaps of the two real positions are not silently assumed to preserve the kernel.

P01 admits the tested interchange. In P02 the unequal row tensors break it, and the false symmetry is correctly rejected. P03 and P04 likewise do not obtain the P01 orbit reduction. Discovering that a symmetry is absent can still cost time. This explains why certified symmetry is an opportunity, not an unconditional speed guarantee.

6 The deeper-recursion test and its exact checks

6.1 Three active levels, not merely three evaluations

In D01 the active depths are zero, one, and two, with leaves at depth three. Every child is strictly smaller than its parent, while total child ports sum to twice the parent's edge count. The evaluator uses the returned child contractions to construct local polynomials and recover the parent target. Thus the deeper test is not a label assigned to a nonrecursive interpolation calculation.

A tensor created at precision p_c carries fixed canonical coefficient representatives thereafter. When a later call requests more precision, the old tensor is not symbolically re-created from its ancestor. Otherwise the polynomial being recursively evaluated could change. Creation data are included in cache keys and symmetry certificates. Descendant symmetry is checked afresh; a root automorphism is not automatically inherited without verifying the new tensors.

6.2 Measured structural work

Diagnostic counters for one D01 execution are:

Quantity	S	D	G	DG
Active calls	54913	54913	4303	4303
Leaf calls	981032	981032	75790	75790
Child requests	12648064	12648064	886168	886168
Cache hits	11612120	11612120	806076	806076
Nonbase active returns	20892	20892	3142	3142

Child requests include cache hits; active and leaf calls count executed contractions. The equal structural counts within S/D and within G/DG show that the precision change does not reduce these counts on this fixture. It primarily changes arithmetic size. Symmetry, in contrast, removes many child requests and executed descendant calls. These exact counts are more structurally informative than a timing ratio alone.

The 3,142 nonbase active returns for DG establish that recursion is not limited to scalar intermediates. The diagnostic execution compares every one of its 4,303 active targets with a direct ring-valued contraction. S and D similarly check all 54,913 executed active targets in their diagnostic runs. This is deeper validation than checking only the root answer one. It is still validation of one factorized deep fixture, not evidence over a broad distribution of hard instances.

7 Why the precision reductions are valid

7.1 The target functional, not just a guessed bit budget

For distinct odd centers t, scale $\alpha = 2L$, and multiplicities q_t, let $Q = \sum_t q_t$ and

$$E_t(T) = \prod_{s \neq t} \left(\frac{T-s}{t-s} \right)^{q_s}, \qquad a_{tk} = [z^k] E_t(t+\alpha z)^{-1}.$$

For $\deg F < Q$, put $c_{tj} = [z^j] F(t+\alpha z)$. Hermite interpolation gives the exact target formula

$$F(0) = \sum_t \sum_{j=0}^{q_t-1} w_{tj} c_{tj}, \tag{3}$$
$$w_{tj} = E_t(0) \sum_{k=0}^{q_t-1-j} a_{tk} (-t/\alpha)^{j+k}.$$

To see why, form

$$H_{tj}(T) = E_t(T) \left(\frac{T-t}{\alpha} \right)^j \sum_{k=0}^{q_t-1-j} a_{tk} \left(\frac{T-t}{\alpha} \right)^k.$$

Its jet at t is z^j through the required order, and its prescribed jets at other centers vanish. It has degree at most $Q-1$. Consequently $F - \sum c_{tj} H_{tj}$ has Q zeros counting multiplicity and degree less than Q, so is zero. Evaluating at zero proves (3).

If the inputs c_{tj} are two-adically integral and are known modulo 2^{p+e}, it suffices that

$$e \geq \max\bigl(0, \max_{w_{tj} \neq 0} -v_2(w_{tj})\bigr)$$

to recover the target modulo 2^p. Each possible input error is a multiple of 2^{p+e}, and multiplying by its target weight leaves a multiple of 2^p. This proof applies coefficientwise in the fixed extension algebra. It is a certificate for a specified functional on a specified polynomial class, not a universal minimum precision for all algorithms on the underlying counting instances.

7.2 Separating conservative slack from degree information

The finite source-form schedule uses $q_t = 3M$ and

$$e_S = 24ML + 8M(a+1), \qquad a = \log_2 L.$$

The degree-aware map instead uses balanced multiplicities totaling $D+1$, where $D = M(2L-1)$, and its exact denominator valuation.

L	M	D	S increment used	Exact loss of S weights	D increment used
4	4	28	480	81	49
2	4	12	256	34	18
2	3	9	192	25	13
2	2	6	128	16	9

The 480-to-49 comparison combines two changes: replacing a conservative source schedule by actual target losses and reducing the required Hermite class using the known degree. The unchanged source target map already has exact loss 81, not 480. There is no timed variant isolating that 480-to-81 change, so one cannot attribute the entire measured speedup solely to the smaller polynomial class or to a uniquely Ram Calculus mechanism.

For the packet degree map, the multiplicities are $(8, 7, 7, 7)$ at centers $(-3, -1, 1, 3)$, totaling 29. S has 48 formal Hermite channels. Nevertheless both implementations still recover all 29 coefficients at each center using 116 local evaluations. V6 does not demonstrate a speedup from deleting those interpolation evaluations.

7.3 Actual working precision and the guard bits

Packet case	Output p	S/G coefficient bits	D/DG coefficient bits	Numerator guard bits added
P01–P03	5	485	54	4
P04	65	545	114	4

The 485-to-54 change is about an 88.9% reduction in the coefficient bit budget; 545-to-114 is about 79.1%. Neither percentage is a measured reduction in total memory or runtime.

The factor $\mathcal{C}'(0) = -144 = 16(-9)$ requires special care. At $T = t + 8\zeta$, four factors of $\mathcal{C}$ are units. Of the four odd-root factors, one has valuation at least three and the others have valuations one, one, and two. Thus $\mathcal{C}(T)$ is divisible by 2^7. Dividing by 16 leaves at least three powers of two, and division by -9 is multiplication by a unit inverse. Computing the numerator modulo 2^{B+4} protects the exact division and yields the quotient modulo 2^B.

Hence Boolean packet numerator calculations use 489 versus 58 bits; the weighted calculation uses 549 versus 118 bits. The working-bit counter does not bound every temporary integer or rational numerator in the implementation. In particular, coefficients with an odd denominator are valid elements of $\mathbb{Z}_2$, even when they are not ordinary integers. The independent oracle retains exact fractions and reduces n/d as nd^{-1} modulo 2^B when d is odd.

7.4 The recursive precision paths

The deeper path follows the successive active sizes $4, 3, 2$:

$$\begin{aligned} S/G &: 5 \longrightarrow 261 \longrightarrow 453 \longrightarrow 581, \\ D/DG &: 5 \longrightarrow 23 \longrightarrow 36 \longrightarrow 45. \end{aligned}$$

The maximum coefficient budget drops from 581 to 45 bits, about 92.3%. These are executed paths, not an extrapolation obtained by multiplying a one-level saving by a guessed number of levels. Preserving creation-time lifts remains essential while requesting these different descendant precisions.

8 Exact validation and the measurement protocol

8.1 What passed

Five methods on five fixtures, with three timing repeats and one diagnostic execution per method/fixture, give

$$5 \cdot 5 \cdot 3 = 75 \text{ timing workers}, \qquad 5 \cdot 5 = 25 \text{ diagnostic workers}.$$

All 100 final-suite outputs agree with independent exact root targets. Every non-direct worker must also return the complete local polynomial at every expected center, at its own recorded precision. Root nonbase coordinates are checked and vanish as required. Direct workers have no local polynomial and are not counted as passing a polynomial check.

For packets, the independent oracle constructs K_{xy} by exact rational polynomial products and expands the double Boolean-assignment sum. It does not reuse the geometric-mode construction or recursive evaluator. Diagnostic executions additionally compare the closed kernel and every executed active child target with exact direct calculations. This checks distinct layers of the implementation, although all use the same mathematical problem specification.

The saved auxiliary controls include 31 field inverses, 192 packet coefficient identities, false-symmetry rejection, and the higher-mode omission counterexample. Precision checks include 200 monomial-basis checks, of which 148 are through the engine degree bounds; 126 uncompiled source-form basis comparisons; 90 residue-recovery checks; 30 literal-source residue comparisons; and 84 two-node closed-form checks. There are 52 rejection checks and fixed-map out-of-class and one-bit-short witnesses. Such witnesses concern fixed maps and input classes, not lower bounds for every possible tensor-counting algorithm.

These are deterministic finite checks and proof certificates, not 100 independent random problem instances or a statistical reliability bound.

8.2 The interrupted run and complete timing restart

The original run completed 25 diagnostic workers and 22 timing workers before its first deep source timing worker exceeded a configured 900-second subprocess timeout with a large elapsed-time overrun. Its cause was not established, and it returned no exact result.

The 47 completed records and the interruption evidence were preserved. The *entire* timing phase was restarted with a 3600-second worker timeout and CPU-time recording. The final tables use 75 restarted timing workers and the 25 retained diagnostics. The 22 earlier timings were retired as a whole, not selectively discarded as inconvenient outliers. They are not mixed into the medians. The failed attempt is not counted among the 100 validated outputs.

8.3 Included costs and environmental limits

Workers are sequential fresh processes, with method order rotated across repeats and fixtures. Process startup and imports are outside the timed pipeline interval. Network construction, precision planning, symmetry discovery, geometric summation, child creation, caches, arithmetic, interpolation, and recovery are included. The closed-kernel value is computed as common overhead even when its diagnostic assertion is disabled. Independent oracle construction and diagnostic timing are not included in performance medians.

The environment was CPython 3.12.14 on Windows 11 build 26300, Intel64 Family 6 Model 140 Stepping 1. Three repeats are fewer than v5's seven. CPU time helps distinguish scheduled work from some elapsed-time interruption, but remains sensitive to processor frequency and implementation details. Very short direct calls can have zero or quantized CPU time; their CPU ratios are not reported.

9 Speedups, variability, and failure to improve some cases

9.1 Median runtimes on all five fixtures

All entries below are seconds, including the direct column.

Case	Direct	S	D	G	DG
P01	0.000252	4.216	0.780	3.521	0.556
P02	0.000238	3.455	0.715	3.948	0.732
P03	0.000410	3.095	0.643	3.741	0.652
P04	0.000254	5.013	0.931	4.632	0.875
D01	0.000547	858.467	749.932	32.927	22.094

Ratios are computed from unrounded medians; a value below one means the variant was slower than S in that comparison.

Case	S/D	S/G	S/DG
P01	5.40	1.20	7.58
P02	4.83	0.88	4.72
P03	4.81	0.83	4.75
P04	5.39	1.08	5.73
D01	1.14	26.07	38.85

P01 shows a useful combination of lower precision and certified packet interchange. The combined packet ratios range from 4.72× to 7.58×. In P02 and P03 symmetry alone is slower than S, with ratios 0.88 and 0.83. The absence of useful orbits leaves certification overhead without a compensating structural reduction; machine variation also prevents attributing every timing difference to that overhead.

Adding symmetry to D does not show a consistent incremental benefit on those asymmetric cases: P02/DG has median 0.732 s versus 0.715 s for D, and P03/DG has 0.652 s versus 0.643 s. These small differences are descriptive, not statistically established slowdowns. The safe conclusion is that symmetry is not automatically worthwhile.

9.2 The deeper fixture in detail

Method	Elapsed median	Elapsed minimum	Elapsed maximum	CPU median
S	858.467	422.564	871.824	825.984
D	749.932	329.560	1793.712	649.062
G	32.927	28.972	72.069	32.734
DG	22.094	21.437	22.709	21.703

The combined elapsed-time ratio is 38.85×, and the ratio of median CPU times is 38.06×. G alone is 26.07× faster than S by elapsed medians; DG is another 1.49× faster than G on that comparison. D alone has only a 1.14× elapsed-time ratio. One should not multiply the separate S/D and S/G ratios and call the result an independently established combined benefit.

The active-call counts provide a structural explanation for part of the large difference: S/D execute 54,913 active calls while G/DG execute 4,303. Precision reduces coefficient size, but does not reduce those counts. The experiment does not isolate all causal contributions of arithmetic, cache reuse, symmetry, and operating conditions.

In the second completed D01/D repeat, elapsed time was 1793.712 s while CPU time was 649.062 s. This reveals substantial time outside scheduled computation; it does not identify suspension, scheduling, or another system cause. That completed sample remains in the tables. Even source CPU times range from 418.828 to 830.125 s. Therefore the reported ratios are host-specific descriptions, not stable hardware-independent factors or controlled causal estimates.

9.3 Why none of this establishes practical SAT superiority

Direct counting is faster on *every* fixture. The direct D01 median is about 0.000547 s, compared with 22.094 s for DG. On four variables, constructing many recursive children and interpolating polynomials costs far more than enumerating 16 assignments.

The significance of the speedups is thus local: the reductions save work inside the specified recursive comparison. They do not show that the resulting solver is competitive with direct counting at these sizes, with modern SAT solvers on realistic problems, or with the full source algorithm in its intended asymptotic regime. Packet and deep-family ratios cannot be multiplied, because no measured run combines their most demanding features.

10 Memory: what was measured and what cannot be concluded

10.1 The selected metric

V6 records Windows peak process working set, in MiB, from each untraced timing worker. This includes the interpreter, imports, allocator retention, and other process-resident memory. It also records the process's pre-call high-water mark. This is not the same quantity as v5's traced peak Python allocations, so the two versions' memory numbers cannot be directly compared.

The median peaks are:

Case	Direct	S	D	G	DG
P01	25.82	25.76	25.66	25.79	25.65
P02	25.95	25.91	25.75	26.10	25.88
P03	25.93	25.74	25.71	25.93	25.97
P04	25.86	26.01	25.84	26.04	25.89
D01	25.90	26.02	25.94	26.04	25.81

Every one of the 75 timing workers had exactly zero increase over its own pre-call process high-water mark. This is an equality in the saved byte counts, not merely a zero obtained by rounding MiB. The metric therefore did not resolve algorithm-specific peak use.

10.2 The approximately 0.8 percent difference is not a saving

D01/S and D01/DG have median peaks of 26.02 and 25.81 MiB. Their approximately -0.8% relative difference reflects pre-call process baselines. It is *not evidence that DG reduces algorithm memory by 0.8%.* Nor does a zero high-water increase mean that the algorithm allocated no memory: it may allocate and release objects entirely below a previous peak.

What is established is narrower: the measured calls did not exceed their own process high-water marks. What is not established includes the ordering of algorithm-specific peak memory across methods, a percentage memory saving, an optimal workspace bound, or preservation of the source's claimed asymptotic space bound.

Smaller coefficients can reduce the storage of individual arithmetic objects, while caches, orbit data, interpolation bases, and interpreter overhead can dominate total storage. The precision theorem is therefore not a memory theorem. A dedicated measurement would need isolated allocation profiling or suitably controlled in-call resident/private memory observation, with profiler overhead separated from runtime and cache/certificate storage included. This is future work, not another measurement already present in v6.

11 Proved improvements, empirical evidence, and Ram Calculus's role

11.1 Results that are justified mathematically

The central finite results are exact moment coalescence on the two-coordinate support; coefficient extraction into two-port child tensors; Hermite target reconstruction on a specified degree class; valuation-certified precision and protected division; and orbit sums under explicit invariance certificates. Their proofs do not depend on the measured runtime ratios.

The independent exact-oracle checks, active-child checks, and negative controls then provide evidence that these identities were implemented correctly on the tested families. They do not make the families general or replace a proof about every input to the full algorithm.

11.2 What is not uniquely Ram Calculus

The Ram Calculus interpretation organizes the method around an explicit observation space, quotient by redundant observations, target functional, and precision certificate. It makes the representation choice and its admissible scope visible and testable.

The mechanisms are also recognizable conventionally: equality of finite-support moments, Boolean polynomial reduction, Hermite interpolation, valuation arithmetic, and certified group-orbit sums. A conventional implementation using the same identities would obtain the same exactness and potential performance benefits. V6 does not demonstrate an advantage exclusive to the name Ram Calculus, a new complexity class separation, or historical priority for these classical ingredients. Its contribution is their explicit integration, local certificates, tests, and carefully delimited extensions.

11.3 Advantages and costs

The advantages are transparent conditions for recovery, less conservative arithmetic budgets, a visible reduction of redundant mode data, and certified reuse rather than guessed symmetry. The

costs include algebraic construction, certificate computation, child-network management, caching, and sensitivity to the exact input class. Small direct problems can be made much slower by this machinery. Asymmetric inputs may offer no useful orbits, and larger support sets may not permit the same moment quotient.

12 Further exact algebra and unmeasured opportunities

12.1 A six-state to four-state Boolean quotient

V6 proves an additional algebraic simplification that is not used by the timed code. Write a row polynomial as

$$c_{00} + c_{10}Z_0 + c_{01}Z_+ + c_{20}Z_0^2 + c_{11}Z_0Z_+ + c_{02}Z_+^2.$$

For Boolean $x = X_0$, $y = X_2$, substitute $Z_0 = x + y$, $Z_+ = y$ and use $x^2 = x$, $y^2 = y$. The result is

$$\begin{aligned} &c_{00} + (c_{10} + c_{20})x \\ &\quad + (c_{10} + c_{01} + c_{20} + c_{11} + c_{02})y + (2c_{20} + c_{11})xy. \end{aligned}$$

This identity requires no division. Evaluation of $1, x, y, xy$ on the four Boolean points gives a unimodular matrix, so these observations form a basis over the modular rings in question, despite their zero divisors. Thus four states suffice algebraically instead of six.

Two packets would then have $4^2 = 16$ assignments, rather than 36; under certified interchange, ten unordered representatives would replace 21. These are proved *state counts*, not observed runtime or memory savings. The measured implementation retains six states. New child tensors, certificates, and cache behavior must be implemented and tested before claiming an end-to-end improvement.

12.2 Other concrete next experiments

1. Replace the geometric loop by the closed expression for B_x in a separately timed variant, isolating resummation from precision.
2. Keep the source target weights but use their exact denominator loss, separating conservative-schedule slack from degree-class savings.
3. Implement the four-state Boolean quotient and measure all construction, symmetry, cache, and allocation costs.
4. Test at least three distinguishable tile coordinates, where higher moments need not coincide, with explicit rank certificates.
5. Combine multi-port children with deeper recursion on connected networks, rather than combining two separate families only verbally.
6. Investigate selective-channel reconstruction and streaming evaluation with complete error, bit-cost, and space certificates.

Each would be a new experiment or theorem. None is already established as a full-algorithm performance gain by the v6 measurements.

13 What remains necessary for the full algorithm

13.1 The literal activation barrier

The retained source analysis uses

$$\lambda = \lceil \log_2 s \rceil, \qquad u_0 = \left\lfloor \frac{\lambda}{131072} \right\rfloor .$$

A positive root parameter requires $\lambda \geq 131072$, hence $s \geq 2^{131071} + 1$, a 39,457-digit integer. This concerns the magnitude of the instance-size parameter, not the storage needed to write its binary representation. The source leaf threshold also grows with s, so declaring an enormous s does not activate a four-edge input. The local finite tests therefore rescale parameters and leaf thresholds; they do not execute the paper's literal active regime.

Large constants alone do not disprove an asymptotic theorem. Conversely, successful small rescaled identities do not verify that theorem's entire parameter schedule, reductions, correctness argument, or complexity accounting. This summary does not update the literature status of the source's headline ETH assertion.

13.2 Requirements for a justified algorithmic improvement

A full result would require the following, not just a faster local kernel on five fixtures:

1. **Uniform domain coverage:** prove appropriate observation, degree, and reconstruction certificates for the actual packet families, including mixed local/crossing behavior and larger moment spaces.
2. **Recursive closure:** show that every generated child stays within the certified class, with valid creation-time lifts, shrinkage, and precision at all depths.
3. **Worst-case symmetry accounting:** include the cost of discovering and representing groups and handle instances with no useful symmetry. Favorable root automorphisms are not a worst-case guarantee.
4. **Complete bit complexity:** charge coefficient arithmetic, extension construction, local evaluations, target-map construction, child tensors, and all certification, not just state counts.
5. **Space preservation:** account for caches, interpolation bases, group data, and recursion storage in a streaming or otherwise space-controlled implementation.
6. **Full pipeline comparison:** integrate the reductions into the relevant counting/SAT reduction and solve the resulting recurrence. Only then could a better general bound be asserted.
7. **Stronger empirical evidence:** use larger and connected families, genuinely independent higher moments, combined depth and packet complexity, and better controlled time and memory measurements.

No observed v6 ratio supplies these missing quantifiers. In particular, the packet symmetry count 21, the proposed count ten, and the deep speed ratio 38.85× cannot be substituted into a general SAT exponent without a uniform argument.

14 Overall assessment and evidence trail

The defensible conclusion is constructive but limited. V6 demonstrates that exact observation reduction, degree-aware precision, and certified symmetry can coexist with nonzero higher-mode calculations and actual recursive children. It broadens the finite validation beyond v5 and shows substantial host-specific speedups against the finite comparator. It also exposes failures of unjustified mode omission, unprofitable symmetry cases, and the inadequacy of a startup-dominated process peak for claiming memory improvements.

The strongest new result is not a claim to have improved the general SAT bound. It is a documented set of exact local reductions, precision certificates, and matched tests, with a precise list of what must still be proved and implemented. Multi-edge depth, large independent moment systems, and full time/space improvement remain open in this investigation.

The principal local sources are [1, 2]. The completed evidence directory is

`eth-benchmark-v6/results-20261027-resumed/`.

It contains `results.json`, `workers.jsonl`, `instances-and-oracles.json`, `controls.json`, and `recovery-provenance.json`. The interrupted original evidence is preserved in `eth-benchmark-v6/results-20261027/`. Implementations are `recursive_v6.py`, `precision_v6.py`, `run_v6.py`, and the recovery driver `resume_v6.py`. Frozen v4/v5 modules and the pinned source are read-only dependencies. Hash registers distinguish scientific data from the separately authorized author-only updates to old reports.

This summary adds no benchmark workers and alters no prior report or raw result. Its tables and arithmetic are checked against the saved evidence. Publication formatting checks are limited to at most two iterations; minor remaining typographical warnings are not mathematical claims or experimental failures.

References

[1] M. Subbarao. *Ram Calculus and the Exponential Time Hypothesis, Revision 6: Multi-Edge Packets, Nonzero Higher Modes, and Three Active Recursive Levels*. Technical report, October 27, 2026, 80 pages. Local PDF; `Ram-Calculus-Exponential-Time-Hypothesis-v6.tex`. New results: Section 1, physical PDF pages 6–19.

[2] M. Subbarao. *ETH benchmark v6: finite packet and deeper-recursion evidence*. `eth-benchmark-v6/results-20261027-resumed/`, October 27, 2026. Exact fixtures, local polynomial oracles, worker records, recovery provenance, and source hashes. No new experiment is reported in this summary.

[3] OpenAI. *The Exponential Time Hypothesis is False*. October 17, revised October 25, 2026. Source snapshot cited by v6: `42907c9e999cc00d6abe7a50bfc1023250edd1b9`. https://github.com/openai/math/tree/42907c9e999cc00d6abe7a50bfc1023250edd1b9/preprints/The-Exponential-Time-Hypothesis-is-False-October-25-2026. The title identifies the source's claim, not an independent conclusion established by this summary.